

Yuezhou Hu

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Research Experience

PhD Student , Berkeley Artificial Intelligence Research Lab (BAIR), University of California, Berkeley, Advisor: Prof. Kurt Keutzer	Sep. 2025 – Present
Research Intern , H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology, Advisor: Prof. Tuo Zhao	Jul. 2024 – Aug. 2024
Research Intern , Statistical Artificial Intelligence & Learning Group, Tsinghua University, Advisor: Prof. Jianfei Chen , Prof. Jun Zhu	May 2022 – Jul. 2025

Education

University of California, Berkeley , Computer Science Ph.D Student Department of Electrical Engineering and Computer Sciences	Sep. 2025 – Present
Tsinghua University , B.S. in Computer Science Department of Computer Science and Technology	Sep. 2021 – Jul. 2025

Research Interests

My research interests primarily focus on **efficient machine learning**, particularly efficient training and inference. Recently, I am interested in:

- Diffusion Language Model; Speculative Decoding; Large Reasoning Model; Dynamic Sparse Training

Publications

4 first/cofirst-author papers, *equal contribution

- Residual Context Diffusion Language Models** [\[arXiv\]](#) [\[Code\]](#)
Yuezhou Hu^{*}, Harman Singh^{*}, Monishwaran Maheswaran^{*}, Haocheng Xi, Coleman Hooper, Jintao Zhang, Aditya Tomar, Michael W. Mahoney, Sewon Min, Mehrdad Farajtabar, Kurt Keutzer, Amir Gholami, Chenfeng Xu
International Conference on Machine Learning (ICML), 2026
- ParallelBench: Understanding the Trade-offs of Parallel Decoding in Diffusion LLMs** [\[arXiv\]](#)
Wonjun Kang, Kevin Galim, Seunghyuk Oh, Minjae Lee, Yuchen Zeng, Shuibai Zhang, Coleman Hooper, *Yuezhou Hu*, Hyung Il Koo, Nam Ik Cho, Kangwook Lee
International Conference on Learning Representations (ICLR), 2026
- AdaSPEC: Selective Knowledge Distillation for Efficient Speculative Decoders** [\[arXiv\]](#) [\[OpenReview\]](#)
Yuezhou Hu^{*}, Jiaxin Guo^{*}, Xinyu Feng, Tuo Zhao
Neural Information Processing Systems (NeurIPS), 2025 (**Spotlight**)
- Pruning Large Language Models with Semi-Structural Adaptive Sparse Training** [\[arXiv\]](#) [\[Project page\]](#)
Weiyu Huang, *Yuezhou Hu*, Guohao Jian, Jun Zhu, Jianfei Chen
AAAI Conference on Artificial Intelligence (AAAI), 2025
- S-STE: Continuous Pruning Function for Efficient 2:4 Sparse Pre-training** [\[arXiv\]](#) [\[OpenReview\]](#) [\[Project page\]](#)
Yuezhou Hu, Jun Zhu, Jianfei Chen
Neural Information Processing Systems (NeurIPS), 2024
- Accelerating Transformer Pre-training with 2:4 Sparsity** [\[arXiv\]](#) [\[OpenReview\]](#) [\[PDF\]](#) [\[Project page\]](#)
Yuezhou Hu, Kang Zhao, Weiyu Huang, Jianfei Chen, Jun Zhu
International Conference on Machine Learning (ICML), 2024

Preprints

- **Arbitrage: Efficient Reasoning via Advantage-Aware Speculation** [\[arXiv\]](#) [\[Project page\]](#)
Monishwaran Maheswaran^{*}, Rishabh Tiwari^{*}, ***Yuezhou Hu***^{*}, Kerem Dilmen, Coleman Hooper, Haocheng Xi, Nicholas Lee, Mehrdad Farajtabar, Michael Mahoney, Kurt Keutzer, Amir Gholami
- **A Survey of Efficient Attention Methods: Hardware-efficient, Sparse, Compact, and Linear Attention**
Jintao Zhang, Rundong Su, Chunyu Liu, Jia Wei, Ziteng Wang, Pengle Zhang, Haoxu Wang, Huiqiang Jiang, Haofeng Huang, Chendong Xiang, Haocheng Xi, Shuo Yang, Xingyang Li, ***Yuezhou Hu***, Tianyu Fu, Tianchen Zhao, Yicheng Zhang, Boqun Cao, Youhe Jiang, Chang Chen, Kai Jiang, Huayu Chen, Min Zhao, Xiaoming Xu, Yi Wu, Fan Bao, Jun Zhu, Jianfei Chen
- **CAST: Continuous and Differentiable Semi-Structured Sparsity-Aware Training for Large Language Models** [\[arXiv\]](#)
Weiyu Huang, ***Yuezhou Hu***, Jun Zhu, Jianfei Chen

Technical Skills

Deep learning programming: Python, Pytorch

GPU Programming: OpenAI Triton, C++ , CUDA

Others: Docker, Django, Rust

Language Skills: TOEFL 104 (R:28/L:27/S:23/W:26)

Honors

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|---|-----------|
| • Tsinghua Academic Preeminence Scholarship | Fall 2024 |
| • 84 Future Innovation Scholarship | Fall 2024 |